

RECONPULSE // v1.0

Automated Reconnaissance & Footprinting Utility | Security Engineering Portfolio

DEVELOPER

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PRIMARY LANGUAGE

Python 3

TARGET STACK

Nmap, Requests, CLI

PROJECT ABSTRACT & OBJECTIVE

ReconPulse is a lightweight, modular security tool engineered to streamline initial target discovery and footprinting phases during authorized assessments. By programmatically combining DNS resolution, HTTP header security auditing, and native Nmap subprocess execution, it minimizes manual overhead and standardizes early intelligence gathering.

DEVELOPMENT ENVIRONMENT & STRUCTURE

Built inside a Linux-native development workflow, the script leverages robust error handling, modular functions, and standard libraries to guarantee stability across virtualized testing environments.



Figure 1: Modular script development and structuring inside VS Code workspace.

CORE ENGINEERING MODULES

- **Target Resolution:** Maps target domains to IPv4 addresses using socket verification with exception handling for unreachable hosts.
- **HTTP Header Inspection:** Automatically probes HTTPS endpoints with fallback mechanisms to capture response statuses and review security headers.
- **Service Enumeration:** Integrates optimized Nmap flags (`-T4 -F -sV`) to profile top ports and service versions.

EXECUTION & LIVE TERMINAL OUTPUT

Below is a snapshot and log of ReconPulse executing live against an authorized test environment (scanme.nmap.org), illustrating automated target resolution, HTTP header status checks, and Nmap service version detection.



```
(ubuntu@ubuntu:~)$ python3 recon.py -t scanme.nmap.org
RECONPULS

[+] Automated Recon & Footprinting Utility v1.0
[+] Target Resolved: scanme.nmap.org -> 45.33.32.156
--- [ HTTP Headers & Security Audit ] ---
[!] HTTPS connection failed or timed out for https://scanme.nmap.org. Trying HTTP...
[+] HTTP Status Code: 200
--- [ Nmap Port Scan & Service Detection ] ---
[+] Running basic service scan on top common ports...
Starting Nmap /usr/bin/nmap ( https://nmap.org ) at 2024-09-22 20:38 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.000s latency).
Not shown: 65 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.6.p1 Ubuntu 80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
Service Info: OS: Linux; CPU: x86_64; Linux kernel:

Service detection performed. Please report any incorrect results at https://nmap.org/report/ .
Nmap done: 1 IP address (1 host up) scanned in 16.98 seconds

[+] Recon completed in 8:00:27.069851
(ubuntu@ubuntu:~)$
```

Figure 2: Live execution showing complete scan results and runtime tracking.

```
$ python3 recon.py -t scanme.nmap.org
Automated Recon & Footprinting Utility v1.0 [+] Target Resolved: scanme.nmap.org -> 45.33.32.156 ---
[ HTTP Headers & Security Audit ] --- [!] HTTPS connection failed or timed out. Trying HTTP... [+]
HTTP Status Code: 200 --- [ Nmap Port Scan & Service Detection ] --- PORT STATE SERVICE VERSION
22/tcp open  ssh      OpenSSH 8.6.p1 Ubuntu 80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
Service Info: OS: Linux; CPU: x86_64; Linux kernel:
[+] Recon completed in 0:00:27.069851
```

SECURITY APPLICATION & TAKEAWAYS

Designing custom tools like ReconPulse reinforces foundational concepts in network communication, error management, and process automation. It bridges the gap between raw command-line utilities and clean, repeatable software engineering practices.